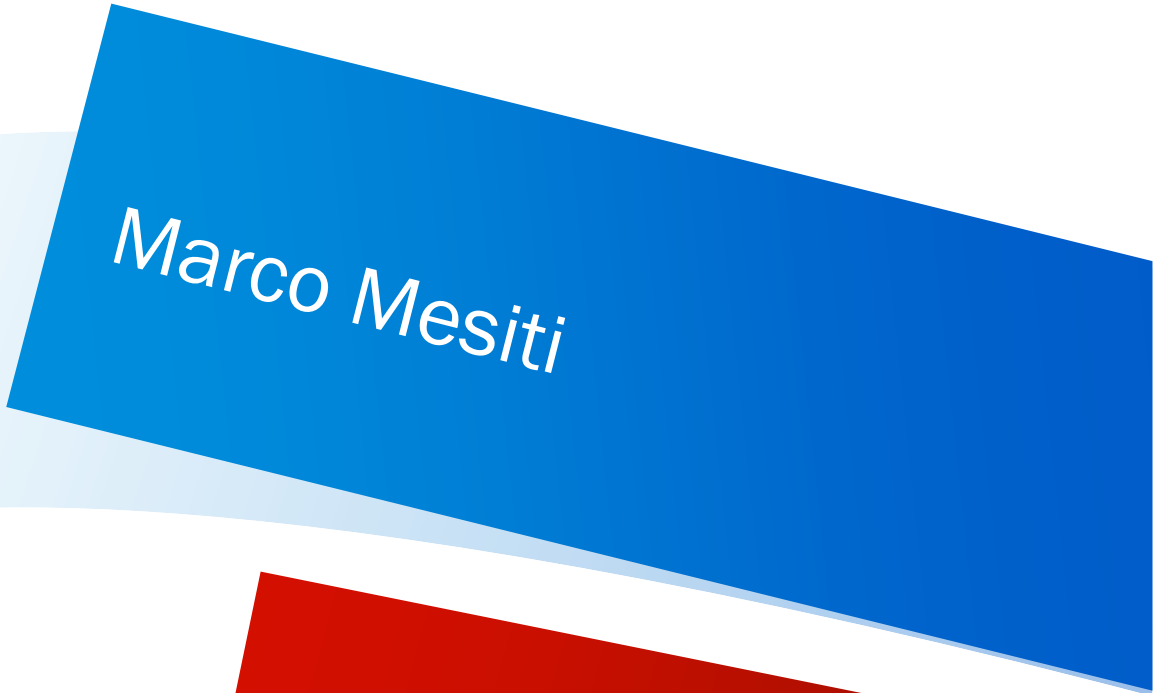
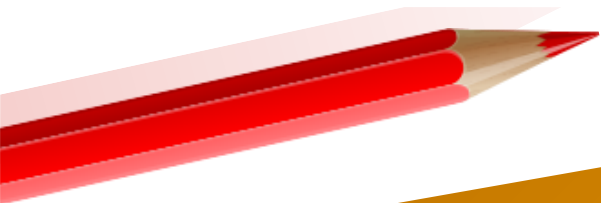




Examples of schema design

Marco Mesiti

Description: Course organization

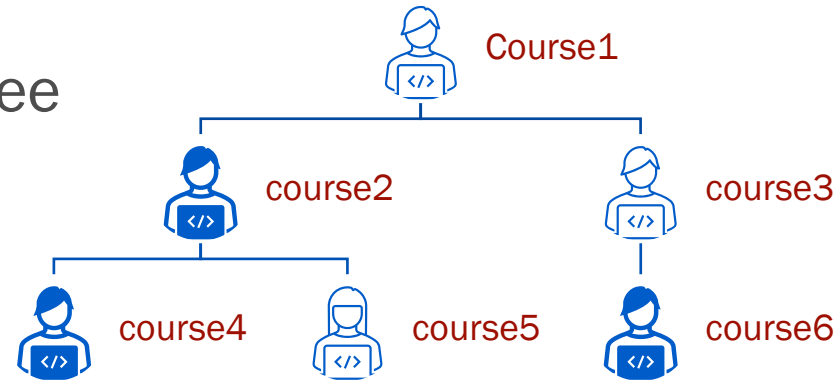


- Consider a big company that organize training courses for its managers. Each course is characterized by a name, a level of difficulty (a number between 1 and 5), a set of topics that are covered during the course, and by at most two preparatory courses (among those offered by the company). Each course is univocally identified by its name. A given course can be attended only if its direct preparatory courses have been already attended. The company may offer more than one edition of each course at different times and venues. Each edition of a course is characterized by a starting date, an ending date, a duration, a venue, a set of teachers (one or more), and a set of attendees (managers of the company), and it is univocally identified by its starting date and venue. Teachers and managers are both employees of the company. Each teacher is characterized by a set of skills and the courses that he/she is able to teach (among those offered by the company). Each attendee is characterized by an initial level of readiness and the final mark that he/she receive at the end of the course. For each manager, besides his/her personal data (name, surname and date of birth), we record his/her education qualification, and previous work experiences. For each previous work experience, we keep track of the type of work done, the name of the company and of its duration.

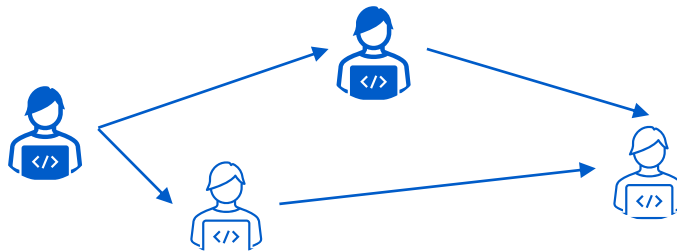
Assumptions

- Assumptions are very important because they allow to disambiguate the textual description of the domain and are the bedrock on which the class diagram is realized
- In this case, the text talk about preparatory courses, but how are they organized?

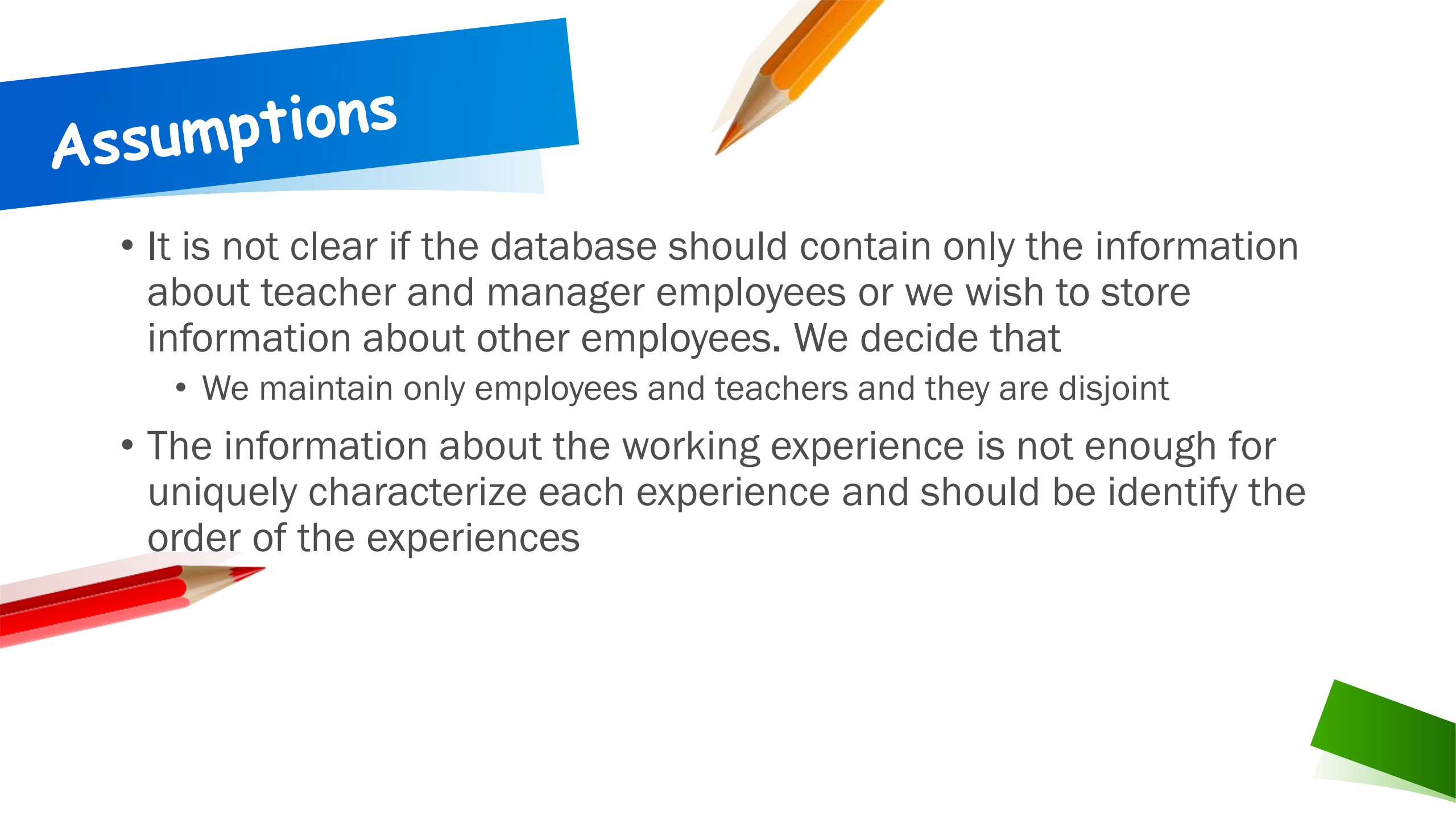
- According to a tree



- according to a DAG

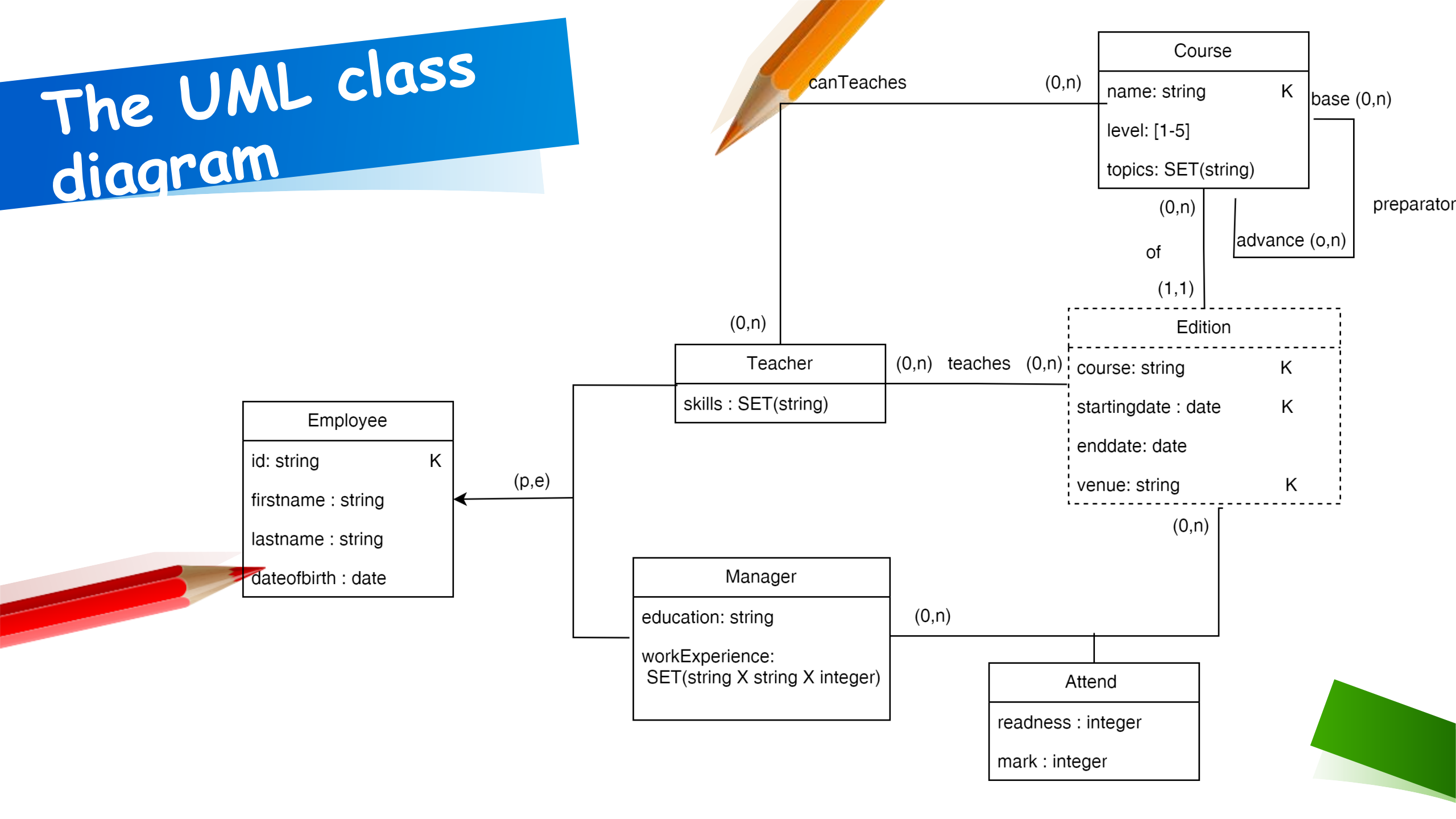


Assumptions



- It is not clear if the database should contain only the information about teacher and manager employees or we wish to store information about other employees. We decide that
 - We maintain only employees and teachers and they are disjoint
- The information about the working experience is not enough for uniquely characterize each experience and should be identify the order of the experiences

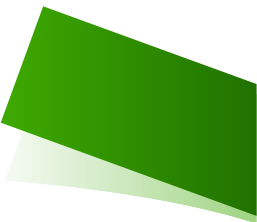
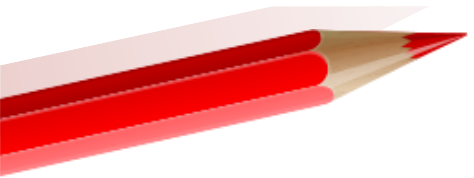
The UML class diagram



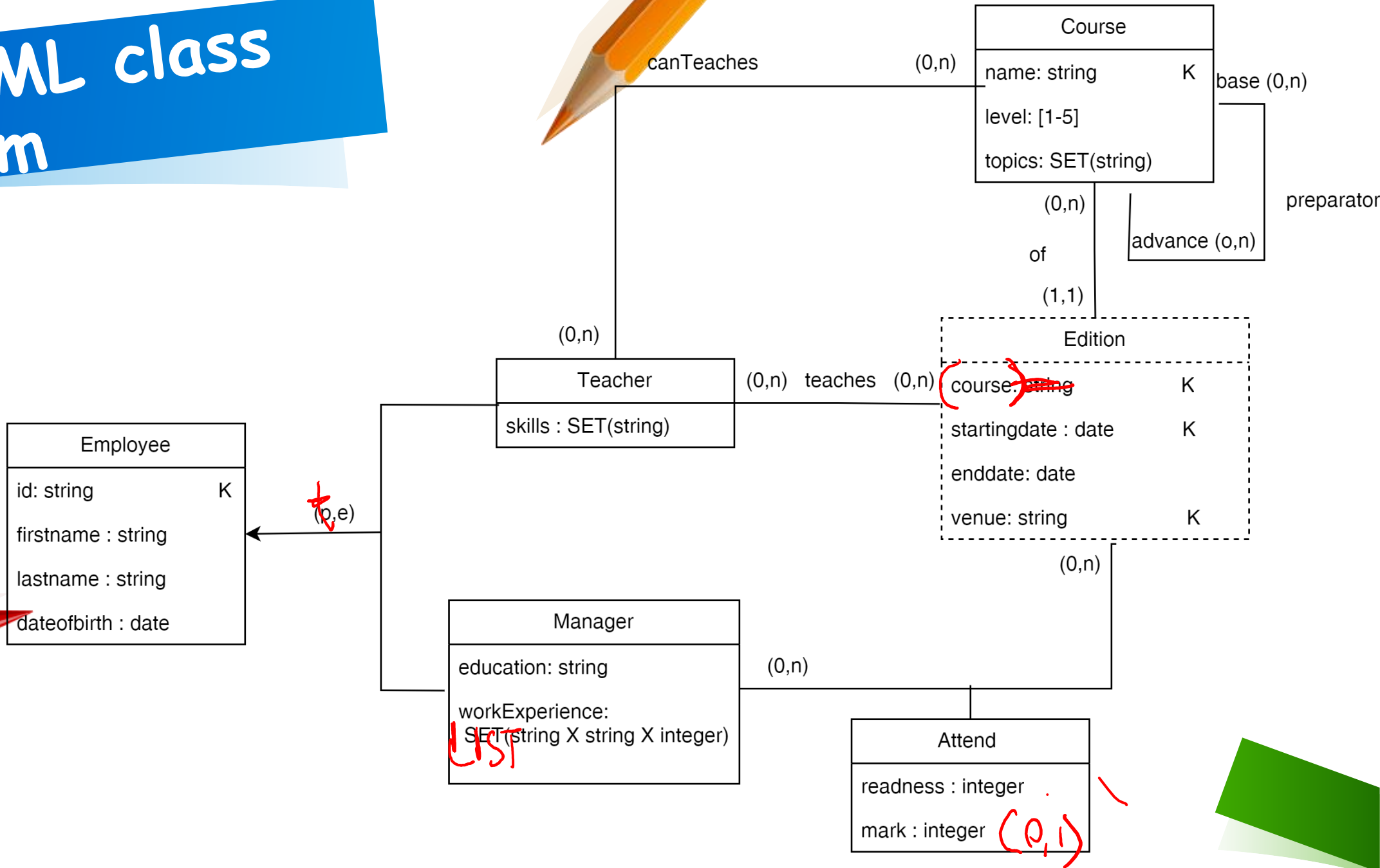
Representation of the workload



- a. Given a course, identify the teachers that can take such course.
- b. Given a course edition, identify the teachers that are assigned to it.
- c. Given a manager, identify if it is allowed to participate to a given course (i.e. he/she has taken part to all the preparatory courses required for the given course).



The UML class diagram



Constraints



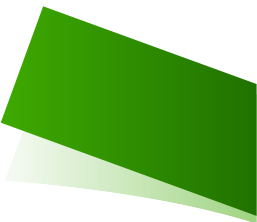
- If a manager m attends an edition e of a course c , and $c_1, \dots, c_n$ are preparatory courses for c , then m should have attended at least one edition for all the courses $c_1, \dots, c_n$
- If a teacher teaches an edition of a course c , he/she should be habilitated to teach the course c
- If a manager attends to editions $e_1, \dots, e_k$ of the same course, his initial level of readiness should be higher each time, i.e.
 $e_i.\text{readiness} < e_j.\text{readiness}$ if $i < j$



Description: management of accidents



- Suppose we wish to develop a class diagram for the management of vehicular accidents occurring on the national road network. Each accident might involve many vehicles (that can be trucks, motorcycles, or cars).
- We are interested in storing the vehicle plate, the length (which is a number between 1 meter and 20 meters), and the date it was first registered. Each vehicle can have onboard one or many individuals that can be the driver or passengers. For each individual, we wish to store the SSN (social security number), first-name and last-name, and age.
- For the drivers, we wish also to store the kind of driving license (“A” for motorcycles, “B” for cars, and “C” for trucks). For each accident we wish to store the network section (e.g. “A1”) and direction (north/south/east/west) where the accident occurred, and the status of the involved individuals when the accident occurred (none, hungry, sleepy, happy, drunkenness). Moreover, a lawyer can be assigned for an accident for which we wish to store the same information of the individuals participating in the accident.
- The workload for this domain is:
 - Given a vehicle identifies the accidents in which it was involved
 - Given an accident, identify the involved participants and driver.
 - Given a lawyer identifies his/her assigned accidents.



UML class diagram

- This design contains errors

